

Calculus II Lesson 02

Integration by Parts Formula: $\int u \cdot dv = u \cdot v - \int v \cdot du$

1) Find $\int x^2 \cdot \ln x dx$

a) Let $u = \ln x$ $\frac{du}{dx} = \underline{\hspace{2cm}} ?$ $du = \underline{\hspace{2cm}} ?$

b) Let $dv = x^2 dx$ $v = \underline{\hspace{2cm}} ?$

c) $\int x^2 \cdot \ln x dx = \underline{\hspace{2cm}} ?$

2) Find $\int x \cdot e^{-2x} dx$

a) Let $u = x$ $\frac{du}{dx} = \underline{\hspace{2cm}} ?$ $du = \underline{\hspace{2cm}} ?$

b) Let $dv = e^{-2x} dx$ $v = \underline{\hspace{2cm}} ?$

c) $\int x \cdot e^{-2x} dx = \underline{\hspace{2cm}} ?$

3) Find $\int x \cdot \sqrt{x-2} dx$

a) Let $u = x$ $\frac{du}{dx} = \underline{\hspace{2cm}} ?$ $du = \underline{\hspace{2cm}} ?$

b) Let $dv = \sqrt{x-2} dx$ $v = \underline{\hspace{2cm}} ?$

c) $\int x \cdot \sqrt{x-2} dx = \underline{\hspace{2cm}} ?$

4) Find $\int x \cdot \cos x dx$

a) Let $u = x$ $\frac{du}{dx} = \underline{\hspace{2cm}} ?$ $du = \underline{\hspace{2cm}} ?$

b) Let $dv = \cos x dx$ $v = \underline{\hspace{2cm}} ?$

c) $\int x \cdot \cos x dx = \underline{\hspace{2cm}} ?$

5) Find $\int x^2 \cdot \sin x dx$

a) Let $u = x^2$ $\frac{du}{dx} = \underline{\hspace{2cm}}?$ $du = \underline{\hspace{2cm}}?$

b) Let $dv = \sin x dx$ $v = \underline{\hspace{2cm}}?$

c) $\int x^2 \cdot \sin x dx = \underline{\hspace{2cm}}?$

6) Find $\int \tan^{-1} x dx$

a) Let $u = \tan^{-1} x$ $\frac{du}{dx} = \underline{\hspace{2cm}}?$ $du = \underline{\hspace{2cm}}?$

b) Let $dv = dx$ $v = \underline{\hspace{2cm}}?$

c) $\int \tan^{-1} x dx = \underline{\hspace{2cm}}?$

7) Find $\int x \cdot \cos 2x dx$

a) Let $u = x$ $\frac{du}{dx} = \underline{\hspace{2cm}} ?$ $du = \underline{\hspace{2cm}} ?$

b) Let $dv = \cos 2x dx$ $v = \underline{\hspace{2cm}} ?$

c) $\int x \cdot \cos 2x dx = \underline{\hspace{2cm}} ?$

8) Find $\int x^2 \cdot e^{2x} dx$

a) Let $u = x^2$ $\frac{du}{dx} = \underline{\hspace{2cm}} ?$ $du = \underline{\hspace{2cm}} ?$

b) Let $dv = e^{2x} dx$ $v = \underline{\hspace{2cm}} ?$

c) $\int x^2 \cdot e^{2x} dx = \underline{\hspace{2cm}} ?$